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 Studierichting: Informatica
 Oefeningenreeks 6B nr. 21.2

$$\begin{aligned}
 & \frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \frac{1}{7 \cdot 10} + \dots + \frac{1}{(3n-2)(3n+1)} \\
 s_n &= \frac{1}{3} \left(1 - \frac{1}{4} + \frac{1}{4} - \frac{1}{7} + \frac{1}{7} - \frac{1}{10} + \dots + \frac{1}{3n-2} - \frac{1}{3n+1} \right) \\
 &= \frac{1}{3} \left(1 - \cancel{\frac{1}{4}} + \cancel{\frac{1}{4}} - \cancel{\frac{1}{7}} + \cancel{\frac{1}{7}} - \cancel{\frac{1}{10}} + \dots + \cancel{\frac{1}{3n-2}} - \frac{1}{3n+1} \right) \\
 &= \frac{1}{3} \left(1 - \frac{1}{3n+1} \right) \\
 &= \frac{1}{3} \left(\frac{3n+1}{3n+1} - \frac{1}{3n+1} \right) \\
 &= \frac{1}{3} \cdot \frac{3n}{3n+1} \\
 &= \frac{n}{3n+1}
 \end{aligned}$$

References